

Nick's Question 0.1 Lemma B.3 Questions

1. On the first line of the proof of Theorem 1.3 of [1] references the bounds from Lemma B.3. My question is about the first inequality right after equation (B.11). I will underline the parts of the inequality that this question pertains to. Here is the inequality:

$$\begin{aligned}
 & |\mathcal{F}(g_{\mathbf{t},t+h,x}^{(n)} - g_{\mathbf{t},t,x}^{(n)})(\xi_1, \dots, \xi_n)|^2 \\
 & \leq \lambda^{2n} \underbrace{J_+^2(t, x)}_{(1)} \underbrace{\prod_{k=1}^{n-1} \exp \left(-\frac{t_{\rho(k+1)} - t_{\rho(k)}}{t_{\rho(k)} t_{\rho(k+1)}} \left| \sum_{j=1}^k t_{\rho(j)} \xi_{\rho(j)} \right|^2 \right)}_{(2)} \\
 & \left| \exp \left(-\frac{1}{2} \frac{t+h - t_{\rho(n)}}{t_{\rho(n)}(t+h)} \left| \sum_{j=1}^n t_j \xi_j \right|^2 \right) - \exp \left(-\frac{1}{2} \frac{t - t_{\rho(n)}}{t_{\rho(n)} t} \left| \sum_{j=1}^n t_j \xi_j \right|^2 \right) \right|^2
 \end{aligned}$$

and it says that this follows from equation (3.5) which says that

$$\mathcal{F}g_{\mathbf{t},t,x}^{(n)}(\xi_1, \dots, \xi_n) = \lambda^n \prod_{k=1}^n \exp \left(-\frac{1}{2} \frac{t_{\rho(k+1)} - t_{\rho(k)}}{t_{\rho(k)} t_{\rho(k+1)}} \left| \sum_{j=1}^k t_{\rho(j)} \xi_{\rho(j)} \right|^2 \right) \quad (0.1)$$

$$\times \exp \left(-\frac{i}{t} \left(\sum_{j=1}^n t_j \xi_j \right) \cdot x \right) \int_{\mathbb{R}^d} \exp \left(-i \left[\sum_{j=1}^n \left(1 - \frac{t_j}{t} \right) \xi_j \right] \cdot x_0 \right) \underbrace{G(t, x - x_0) u_0(dx_0)}_{(1a)}. \quad (0.2)$$

Underline (1): I do not understand why we can factor out $J_+(t, x)$ which is in the (1) underbrace above. If we calculate $|\mathcal{F}(g_{\mathbf{t},t+h,x}^{(n)} - g_{\mathbf{t},t,x}^{(n)})(\xi_1, \dots, \xi_n)|^2$ by using equation (3.5), then because of the (1a) underbrace above we would have that

$$\begin{aligned}
 |\mathcal{F}(g_{\mathbf{t},t+h,x}^{(n)} - g_{\mathbf{t},t,x}^{(n)})(\xi_1, \dots, \xi_n)|^2 &= \left| \dots \int_{\mathbb{R}^d} \exp \left(-i \left[\sum_{j=1}^n \left(1 - \frac{t_j}{t+h} \right) \xi_j \right] \cdot x_0 \right) G(t+h, x - x_0) u_0(dx_0) \right. \\
 &\quad \left. - \dots \int_{\mathbb{R}^d} \exp \left(-i \left[\sum_{j=1}^n \left(1 - \frac{t_j}{t} \right) \xi_j \right] \cdot x_0 \right) G(t, x - x_0) u_0(dx_0) \right|^2
 \end{aligned}$$

and so I do not see why we can factor this out as $J_+(t, x)$ because the integrand of the first term has $(t+h)$ while the second only has (t) .

Relation to Raluca's paper [2]: The corresponding result is found in Lemma 7.2 [2]. The proof is exactly the same up until Raluca's equation (80). However in equation (81), you can see that Raluca took a different approach to bound $\psi_{t,h,x}^{(n)}(\mathbf{t}, \mathbf{t})$ and in return the $A_n(t, h, x)$ function.

Underline (2): Why are we able to factor as was done in this step. Shouldn't we say that

$$|\mathcal{F}(g_{\mathbf{t},t+h,x}^{(n)} - g_{\mathbf{t},t,x}^{(n)})(\xi_1, \dots, \xi_n)|^2$$

$$= \left| \lambda^n \prod_{k=1}^n \exp \left(-\frac{1}{2} \frac{t_{\rho(k+1)} - t_{\rho(k)}}{t_{\rho(k)} t_{\rho(k+1)}} \left| \sum_{j=1}^k t_{\rho(j)} \xi_{\rho(j)} \right|^2 \right) \cdots \right. \\ \left. - \lambda^n \prod_{k=1}^n \exp \left(-\frac{1}{2} \frac{t_{\sigma(k+1)} - t_{\sigma(k)}}{t_{\sigma(k)} t_{\sigma(k+1)}} \left| \sum_{j=1}^k t_{\sigma(j)} \xi_{\sigma(j)} \right|^2 \right) \cdots \right|^2.$$

In other words, the permutations are different so we can't factor like is done to get the first inequality after equation (B.11).

2. Similar to the last question, when calculating $|\mathcal{F}g_{\mathbf{t},t+h,x}^{(n)}(\xi_1, \dots, \xi_n)|^2$ right under equation (B.15), why is there a $J_+^2(t, x)$ term instead of a $J_+^2(t + h, x)$ term?
3. Right after equation (B.16), should $D_{t,h}$ be equal to

$$D_{t,h} = \cup_{\rho \in S_n} \{(t_1, \dots, t_n); 0 < t_{\rho(1)} < \dots < t_{\rho(n)} < t + h, t_{\rho(1)} > t\}?$$

[Lemma 3.2] Where does the $\sum_{\rho \in S_n}$ go when calculating $\mathcal{F}g_{\mathbf{t},t,x}^{(n)} = n! \mathcal{F}\tilde{f}_n = \sum_{\rho \in S_n} \mathcal{F}f_n$?

Nick's Question 0.2 Lemma 3.2 Where does the $\sum_{\rho \in S_n}$ go when calculating $\mathcal{F}g_{\mathbf{t},t,x}^{(n)} = n! \mathcal{F}\tilde{f}_n = \sum_{\rho \in S_n} \mathcal{F}f_n$? This is equation 3.5.

References

BC2017

- [1] Balan, R. and Le, C. (2017) Parabolic Anderson Model with Space-Time Homogeneous Gaussian Noise and Rough Initial Condition *J Theor. Probab.*

BS2017

- [2] Balan, R. and Song, J. (2017) Hyperbolic Anderson Model with space-time homogeneous Gaussian noise *arXiv:1602.07004v2*